

Figure 4: A step-by-step flowchart used by annotators to annotate any pair of review contradiction; Yes:Contradiction, No:Not Contradiction, CNT:Cannot Decide

Xingwei Tan, Yi Cai, and Changxi Zhu. 2019. [Recognizing conflict opinions in aspect-level sentiment classification with dual attention networks](#). In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing, EMNLP-IJCNLP 2019, Hong Kong, China, November 3-7, 2019*, pages 3424–3429. Association for Computational Linguistics.

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Wenting Xiong. 2013. [Helpfulness-guided review summarization](#). In *Proceedings of the 2013 NAACL HLT Student Research Workshop*, pages 77–83, Atlanta, Georgia. Association for Computational Linguistics.

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Weizhe Yuan, Pengfei Liu, and Graham Neubig. 2021. [Can we automate scientific reviewing?](#) *CoRR*, abs/2102.00176.

A Annotation Guidelines

Recognizing Contradiction in a pair of reviews involves analyzing two pieces of text written by two different reviewers about the same paper. The following steps can be followed:

Step 1: Begin by reading the Title and Abstract of the paper to gain an understanding of its subject matter. It is important to read these sections multiple times to grasp the paper’s main points, such as its motivation, contributions, and other relevant aspects. If necessary, refer to the paper itself or read related material to enhance your understanding.

Step 2: Proceed to read and comprehend the reviews of the paper, focusing on understanding

the viewpoints expressed by the reviewers. Take note of their opinions, arguments, and any specific aspects they highlight.

Step 3: Based on the reviewer comments, their aspect-sentiment analysis, and the pair of reviews, you should categorize them accordingly. If the reviews contradict each other, indicating opposing viewpoints or conflicting statements, mark it as a contradiction (C). If there is no contradiction or the reviews are unrelated, mark them as non-contradiction (N). In cases where you find it difficult to determine if there is a contradiction or not, mark it as CON (confused).

To decide if the reviews are contradictory, ask yourself the following question: "If I were shown two contemporaneous documents one containing each of these passages, would I regard it as very unlikely that both passages could be true at the same time? If so, the two contradict each other." You should be able to state a clear basis for a contradiction. For example, the following are contradictions:

R1. *The motivation of using the method for a very small improvement is not convincing.*

R2. *Overall feedback: I found the paper to be well motivated and the proposed approach to be interesting.*

How to make use of the aspect category while annotation?

Given the assigned aspect category, you should utilize the given category to accurately compare these comments. This will help keep your attention strictly on the relevant aspect and refrain from deviating to other topics mentioned in the review text. For example³-

R1: *However, the results are a visible improvement over JPEG 2000 and I don’t know of any other learned encoding which has been shown to achieve this level of performance.*

R2: *...the comparison to JPEG2000 is unfortunately not that interesting since that codec does not have widespread usage and likely never will.*

In Review 1, the reviewer values the comparison made with JPEG 2000, considering it a suitable benchmark and a positive aspect of the paper. Conversely, the reviewer in Review 2 argues that the comparison with JPEG 2000 is not compelling due to its limited use and suggests a comparison with WebP would be more relevant. These statements present opposing viewpoints on the relevance of comparing with JPEG 2000 in the paper. While

³Paper id of the review: ICLR_2017_2

there are many topics discussed in Review 1 and Review 2, it is crucial to concentrate on the remarks concerning the comparison between JPEG and other potential benchmarks when assessing these reviews.

When comparing reviews, keep in mind that one might discuss the subject matter in broad terms, while the other may focus on specific elements.

Consider this example:

R1: *The work is original.*

R2: *The extension to the partially observable setting is interesting as the proposed form finds a common denominator to multiple estimators, but its underlying idea is not novel.*

In this case, Reviewer 1 comments on the overall originality of the paper, declaring it as novel. On the other hand, Reviewer 2 critiques a specific part of the paper—the extension to the partially observable setting—and declares this particular aspect as not novel. Even though Reviewer 2’s comment is more specific, it contradicts the general assertion made by Reviewer 1 about the paper’s novelty.

It is important to identify these types of examples as contradictions, even if they may seem to operate at different levels of specificity. The broad comment about the paper’s novelty in Review 1 is contradicted by the specific critique in Review 2.

You may find more detailed annotation guidelines in our shared repository.

B More details on Aspect Sentiment Model

MIMLLN for Aspect-Category Sentiment Analysis (ACMIMLLN) operates on the assumption that the sentiment of a mentioned aspect category in a sentence aggregates the sentiments of words that indicate that aspect category. In MIMLLN, words that indicate an aspect category are termed ‘key instances’ of that category. Specifically, ACMIMLLN comprises two components: an attention-based aspect category detection (ACD) classifier and an aspect-category sentiment analysis (ACSA) classifier. Given a sentence, the ACD classifier, as an auxiliary task, assigns weights to the words for each aspect category. These weights signify the likelihood of the words being key instances of their respective aspect categories. The ACSA classifier initially predicts the sentiments of individual words. It then determines the sentence-level sentiment for each aspect category by integrating the respective weights with the word sentiments. The

ACD segment comprises four modules: an embedding layer, an LSTM layer, an attention layer, and an aspect category prediction layer. Similarly, the ACSA segment includes four components: an embedding layer, a multi-layer Bi-LSTM, a word sentiment prediction layer, and an aspect category sentiment prediction layer. In the ACD task, all aspect categories utilize the same embedding and LSTM layers, but they have distinct attention and aspect category prediction layers. For the ACSA task, all aspect categories share the embedding layer, the multi-layer Bi-LSTM, and the word sentiment prediction layer, yet they each have unique aspect category sentiment prediction layers.

C Implementation Details

We implemented our system using PyTorch (Paszke et al., 2019), a deep learning framework, and utilized pre-trained transformer models from Hugging Face⁴. The dataset was randomly split into three parts: 80% for training, 10% for validation, and 10% for testing.

For the Aspect Sentiment Model, we conducted experiments with different network configurations during the validation phase. ASAP-Review dataset contains predictions labelled by an aspect tagger on a human-annotated label. We used the 1,000 human-annotated reviews, maintaining the same random split, to train the MIMLLN classifier. Through these experiments, we determined that a batch size of 16 and a dropout rate of 0.5 for every layer yielded optimal performance. The activation function ReLU was used in our model. We trained the model for 15 epochs, employing a learning rate of 1e-3 and cross-entropy as the loss function. To prevent overfitting, we used the Adam optimizer with a weight decay of 1e-3. For the Reviewer Disagreement Classifier, we trained the models using true Sentiment Disparity Aspect pairs with true aspect and sentiment labels. We use a batch size of 16, a maximum length of 280 tokens, and a dropout probability of 0.1. The Adam optimizer was employed with a learning rate of 1e-5. All models were trained on an NVIDIA A100 40GB GPU.

D Result of Aspect Sentiment Classifier

We present the results of the Aspect Sentiment Model in Table 3. We conducted experiments using various settings of BERT transformer layers, such as SCIBERT (Beltagy et al., 2019). SCIBERT

⁴<https://huggingface.co>

Aspect-Category Model Scores		MOT	CLA	SOU	SUB	MEA	ORI	REP
(Kumar et al., 2021)	$F1_{asp}$	0.62	0.67	0.69	0.70	0.72	0.64	0.55
	$F1_{sent}$	0.69	0.70	0.68	0.71	0.52	0.62	0.59
MIMLLN (BERT)	$F1_{asp}$	0.81	0.88	0.78	0.76	0.83	0.86	0.73
	$F1_{sent}$	0.78	0.86	0.76	0.74	0.81	0.83	0.71
MIMLLN(RoBERTa)	$F1_{asp}$	0.80	0.86	0.77	0.75	0.83	0.86	0.71
	$F1_{sent}$	0.76	0.84	0.74	0.73	0.82	0.81	0.69
MIMLLN(SciBERT)	$F1_{asp}$	0.79	0.86	0.77	0.73	0.82	0.85	0.68
	$F1_{sent}$	0.77	0.85	0.75	0.72	0.81	0.81	0.67
MIMLLN(SPECTRE)	$F1_{asp}$	0.75	0.86	0.77	0.73	0.81	0.86	0.68
	$F1_{sent}$	0.74	0.85	0.76	0.73	0.80	0.84	0.66

Table 3: Performance of all the models and baselines from the experiments. Here, ($F1_{asp}$ represents the aspect-category F1 scores, $F1_{sent}$ represents the sentiment-category F1 scores)

is a pre-trained language model based on BERT and trained on a large-scale, labeled scientific dataset. Additionally, we experiment with SPECTER (Cohan et al., 2020), which is a model designed for learning representations of scientific papers. This model is based on a Transformer language model pre-trained on citations. We found that the model performed optimally with BERT for aspect classification across all aspect categories.

Furthermore, we compared our results with those of an aspect and sentiment multi-task model (Kumar et al., 2021). This model utilizes a shared BERT transformer layer and employs different feed-forward networks on top as task-specific layers. For each aspect category, pretrained BERT performed the best (RoBERTa performed the same for meaningful comparison).

Regarding aspect-based sentiment, BERT performed the best for the aspects of Motivation, Clarity, Substance, and Replicability. However, for the Soundness category, SPECTER performed better. As for the Meaningful Comparison aspect, RoBERTa showed better performance.

E Comparsion with ChatGPT

We conducted a comparative study between our proposed baseline model and ChatGPT and BARD⁵, focusing on multi-turn dialogue inputs. We used the OpenAI API⁶ for the task. To evaluate the effectiveness of our model, we experimented with various prompts, selecting the most effective one. The prompt chosen is particularly adept at extract-

ing contradictions. We took a single review pair at a time to detect the contradiction. The structure of this prompt is shown in the Figure 5

Here, {review1} and {review2} are placeholders that represent the text from a pair of reviews for which we aim to identify contradictions.

During our testing, we identified many false positive cases in the output of ChatGPT. For example, when presented with the reviews (paper id: ICLR_2018_456):

- **R1:** *The argument of biological plausibility is not justified*
- **R2:** *Moreover the biological plausibility that is used as an argument at several places seems to be false advertising in my view*

ChatGPT incorrectly labeled these reviews as contradictions. However, our baseline system accurately identified that these statements do not present a contradiction.

Additionally, ChatGPT sometimes compares two distinct aspects. For example:

R1: *This paper is not well-written.*

R2: *The results are reasonable and significant.*

In this instance, the first reviewer is commenting on the clarity of the paper, while the second reviewer is commenting on the substance of the paper. These represent two different perspectives. ChatGPT occasionally fails in such scenarios. However, our proposed baseline incorporates an intermediate aspect sentiment model, which helps reduce these types of errors.

⁵<https://bard.google.com/>

⁶<https://platform.openai.com/docs/api-reference/>

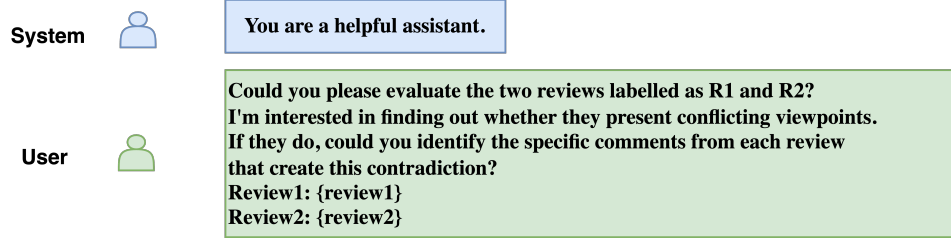


Figure 5: ChatGPT input format

F Error Analysis

We conducted an analysis of the predictions made by our proposed baseline to identify the areas where it most frequently fails.

F.1 Lack of Complete Review Context:

Our proposed baseline can sometimes fail in accurately predicting contradictions when the sentences are significantly related to preceding comments within the same review.

Take, for instance, the following comment: *"This is an elegant intuitive algorithm that, to my knowledge, has not appeared in previous literature."*

In this case, the model incorrectly predicts a contradiction with another review comment that discusses a different algorithm negatively. The issue arises from the model's inability to distinguish between the algorithm discussed in this comment and a different algorithm mentioned negatively by another reviewer. While prediction the model needs to learn from previous discussions as well. Such an approach can be considered as a potential area for future work.

F.2 Issues with Lengthy Comments:

Our proposed baselines sometimes stumble in predicting the right contradictions when dealing with particularly long sentences or complex review comments (significant amount of technical terms or mathematical symbols) which can lead to confusion.

F.3 Error Propagation

We found error propagation from the first model to the second. To illustrate, consider the following example: Reviewer 1's comment reads: "While it is very interesting to apply adversarial noise in real data, this approach is not clearly motivated or explained." The Aspect sentiment model predicts the aspect for this comment as 'soundness' but misses out on 'motivation'. Reviewer 2's comment states: "In overall, I liked its clear motivation

and the simplicity of the method." For this comment, the predicted aspect category by the model is 'motivation'. These two comments provide contrasting viewpoints on the aspect of 'motivation'. Yet, due to the misclassification by the aspect sentiment model, this discrepancy isn't flagged by the second model.